

Program 7:

Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle

Step 1: To build a basic Java Application with Maven Project, run following command:
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMaventoGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
samiksha@samiksha:~/Devops6D$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMaventoGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/samiksha/Devops6D
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMaventoGradle
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/samiksha/Devops6D/MyMaventoGradle
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 6.666 s
[INFO] Finished at: 2026-03-25T11:57:45+05:30
[INFO] -----
```

Step 2: enter the MyMaventoGradle directory and open the pom.xml file.

```
samiksha@samiksha:~/Devops6D$ cd MyMaventoGradle
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ ls
pom.xml  src
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ gedit pom.xml
```

Add the code in the pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-
v4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.example</groupId>

<artifactId>MyMavenApp</artifactId>

<packaging>jar</packaging>

<version>1.0-SNAPSHOT</version>

<name>MyMavenApp</name>

<url>http://maven.apache.org</url>
```

```
<!-- Properties: Customize Java version or plugin versions -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
</properties>
<dependencies>
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>3.8.1</version>
    <scope>test</scope>
  </dependency>
</dependencies>
<!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
```

```

<version>2.22.2</version>

</plugin>

<plugin>

    <groupId>org.apache.maven.plugins</groupId>

    <artifactId>maven-jar-plugin</artifactId>

    <version>3.1.0</version>

    <configuration>

        <archive>

            <manifestEntries>

                <Main-Class>com.example.MainClass</Main-Class>

            </manifestEntries>

        </archive>

    </configuration>

</plugin>

</plugins>

</build>

</project>

```

Step 3 : Run **mvn clean install** to compile your project

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.051 s - in com.example.AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenApp ---
[INFO] Building jar: /home/samiksha/Devops6D/MyMaventoGradle/target/MyMavenApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenApp ---
[INFO] Installing /home/samiksha/Devops6D/MyMaventoGradle/target/MyMavenApp-1.0-SNAPSHOT.jar to /home/samiksha/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/samiksha/Devops6D/MyMaventoGradle/pom.xml to /home/samiksha/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.933 s
[INFO] Finished at: 2026-03-25T12:07:52+05:30
[INFO] -----

```

Step 4: Type the command **mvn exec:java -Dexec.mainClass="com.exempls.App"** to execute

```

samiksha@samiksha:~/Devops6D/MyMaventoGradle$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenApp ---
Hello World!
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.662 s
[INFO] Finished at: 2026-03-25T12:10:12+05:30
[INFO] -----
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ █

```

Step 5: Type the command **gradle init --type pom** to initialize a Gradle Project using gradle init command with type as pom, command converts pom.xml structure to Gradle's build.gradle.

```

samiksha@samiksha:~/Devops6D/MyMaventoGradle$ gradle init --type pom

Select build script DSL:
 1: Kotlin
 2: Groovy
Enter selection (default: Kotlin) [1..2] 2

Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no] no

> Task :init
Maven to Gradle conversion is an incubating feature.
For more information, please refer to https://docs.gradle.org/8.7/userguide/migrating_from_maven.html in the Gradle documentation.

BUILD SUCCESSFUL in 1m 54s
1 actionable task: 1 executed
samiksha@samiksha:~/Devops6D/MyMaventoGradle$

```

Type the ls command to see all the files and directories.

```

samiksha@samiksha:~/Devops6D/MyMaventoGradle$ ls
build.gradle  gradle  gradlew  gradlew.bat  pom.xml  settings.gradle  src  target
samiksha@samiksha:~/Devops6D/MyMaventoGradle$ █

```

Step 6: now open the terminal and type the command **cd MyMavenToGradleApp** to enter the folder and create a **Jenkinsfile**. Paste the code in the Jenkinsfile

```

pipeline {
    agent any // Use any available agent

    tools {
        maven 'Maven' // Ensure this matches the name configured in Jenkins
    }

    stages {
        stage('Checkout') {
            steps {
                git
                    branch: 'master',
                    url: 'https://github.com/Samiksha201104/MyMavenGradleApp.git'
            }
        }
    }
}

```

```

stage('Build') {
    steps {
        sh 'mvn clean package' // Run Maven build
    }
}
stage('Test') {
    steps {
        sh 'mvn test' // Run unit tests
    }
}
stage('Run Application') {
    steps {
        // Start the JAR application
        sh 'java -jar target/MyMavenGradleApp-1.0-SNAPSHOT.jar'
    }
}
}
post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}
}

```

Then commit and push it to GitHub

Step 7:in the Jenkins Dashboard click on new item type the name of the item and select Pipeline

New Item

Enter an item name

MyMavenToGradle

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project

OK

In configure scroll down to Pipeline and choose pipeline script from SCM, give the repository url and choose GitHub Credentials

Jenkins / MyMavenToGradle / Configure

Q ?

Configure

General

Triggers

Pipeline

Advanced

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/Samiksha201104/MyMavenToGradle.git

Credentials ?

GitHub Credentials

+ Add

Advanced

Save Apply

Step 8 : in the Dashboard click on the MyMavenToGradleApp item and select Build Now and then Full Stage view

Correct the errors and build again on jenkins till all the stages are successful



MyMavenToGradle - Stage View

Average stage times: (full run time: ~6min 24s) #3 Apr 12 21:13 No Changes	Declarative: Checkout SCM	Checkout	Build	Test	Run Application	Declarative: Post Actions
	6s	2s	2min 46s	1min 18s	9s	4s
	6s	2s	2min 46s	1min 18s	9s	4s